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Kirobo social robot speaks from International Space Station (Credit: <https://newatlas.com>)

Asus Zenbo is another toy robot that can read stories to children.

Social robots. In future, social robots will play a vital role at home. They can remind elderly people to take their medication and give them company when they feel lonely. Fight against social isolation is the main aim of such social companion robots. Roughly ten per cent of all British people regularly feel isolated. In fact, London has appointed a Minister of Loneliness, in January 2018, and has become the first European nation to do so.

Professor Einstein robot developed by Hanson Robotics can smile, express displeasure and stick his tongue out just as Albert Einstein did. If connected to a tablet or smartphone, Professor Einstein can play brain games or teach the user science. Moreover, it can chit-chat about the weather as well as about famous people.

Presently, PARO robot is being used in more than thirty countries around the world. It is helping patients suffering from diseases like Alzheimer's, Autism, Post Traumatic Stress Disorder (PTSD) and other forms of dementia. PARO has special sensors on its anti-bacterial and soil-resistant fur that responds to touch and makes it blink its eyes.

PARO has a black buttoned nose that has light sensors that help in light detection. The battery lasts about six hours and can be easily recharged. PARO looks at people when it is

addressed, cries in response to stimuli and closes its eyes when it is scratched under the chin.

Kuri, developed by Mayfield Robotics, is a fun friend and helpful assistant. It comes with Wi-Fi, Bluetooth, 1080-pixel camera, and facial recognition and telepresence capabilities. It can talk to other robots, too. It can navigate homes, while avoiding obstacles and pets.

Hardware and software for domestic robots

Challenges and resulting improvements in home robotics technology can be contributed to the simplicity and sensitivity of the whole manufacturing process, which includes both hardware and software. Inexpensive motherboards, such as Arduino and Geekduino, are making home robots affordable and easier to build.

Infrared (IR), GPS sensors, distance sensors, robot actuators, robo controller boards, adaptors, expansion boards, power supply units, light sensors, motor controller boards, resistive flex sensors and 180-degree robot servo motors are essential hardware components used in domestic robots.

Geekduino, by Robot Geek, is a perfect microcontroller (MCU) for a robotic project. Users can easily connect sensors, servos and more to Geekduino. It is completely Arduino-compatible, and can run on the same code as Arduino.